

Problem: $f(x, y) = 3x^2 + 5y^2$, $x^0 = (2, 1)$, $\alpha = 0,1$, $\beta_1 = 0,9$, $\beta_2 = 0,999$, $\epsilon = 10^{-8}$

$$\nabla f(x, y) = (6x, 10y)$$

Update: $m \leftarrow \beta_1 m + (1 - \beta_1) g$, $s \leftarrow \beta_2 s + (1 - \beta_2) g^2$ (elementwise);

$$\hat{m} = m / (1 - \beta_1^{k+1}), \hat{s} = s / (1 - \beta_2^{k+1}), x \leftarrow x - 2\hat{m} / (\sqrt{\hat{s}} + \epsilon); m^0 = s^0 = 0$$

Step $k=0$

$$g^0 = \nabla f(x^0) = (6 \cdot 2, 10 \cdot 1) = (12, 10)$$

$$m^1 = 0,9(0,0) + 0,1(12,10) = (1,2, 1,0)$$

$$s^1 = 0,999(0,0) + 0,001(12^2, 10^2) = 0,001(144, 100) = (0,144, 0,100)$$

$$\hat{m}^1 = \frac{(1,2, 1,0)}{1 - 0,9} = \frac{(1,2, 1,0)}{0,1} = (12, 10)$$

$$\hat{s}^1 = \frac{(0,144, 0,1)}{1 - 0,999} = \frac{(0,144, 0,1)}{0,001} = (144, 100)$$

$$\text{Step}^1 = 2 \frac{\hat{m}^1}{\sqrt{\hat{s}^1} + \epsilon} = 0,1 \left(\frac{12}{\sqrt{144}}; \frac{10}{\sqrt{100}} \right) = 0,1(1,1) = (0,1, 0,1)$$

$$x^1 = x^0 - \text{Step}^1 = (2 - 0,1, 1 - 0,1) = (1,9, 0,9)$$

Step $k=1$

$$g^1 = \nabla f(x^1) = [6 \cdot 1,9; 10 \cdot 0,9] = [11,4; 9,0]$$

$$m^1 = 0,9(1,2; 1) + 0,1(11,4; 9) = [1,08 + 1,14; 0,9 + 0,9] = [2,22; 1,8]$$

$$s^1 = 0,999(0,144; 0,1) + 0,001(11,4^2; 9^2) = [0,143856; 0,0999] + [0,12996; 0,081] = [0,273816; 0,1809]$$

$$\hat{m}^1 = \frac{(2,22; 1,8)}{1 - 0,9^2} = \frac{(2,22; 1,8)}{0,19} = (11,6842; 9,4737)$$

$$s^2 = \frac{(0,273816; 0,1809)}{1 - 0,999^2} = \frac{(0,273816; 0,1809)}{0,001999} = (136,9485; 90,4952)$$

$$\sqrt{s^2} = (11,7039; 9,5129)$$

$$\text{step}^2 = 0,1 \left(\frac{11,6842}{11,7039}; \frac{9,4737}{9,5129} \right) \approx (0,99983; 0,99959)$$

$$x^2 = x^1 - \text{step}^2 \approx (1,9 - 0,99983; 0,9 - 0,99959) \approx (1,8002; 0,8004)$$

K	0		K	1	
	0	1		0	1
g	(12; 10)	(11,4; 9,0)	$\hat{s}$	(144; 100)	(136,98; 90,50)
m	(1,2; 1)	(2,22; 1,8)	step	(0,1; 0,1)	(0,9998; 0,9996)
s	(0,144; 0,1)	(0,2738; 0,1809)	x^{k+1}	(1,9; 0,9)	(1,8002; 0,8004)
$\hat{m}$	(12; 10)	(11,684; 9,474)			

Coordinate-wise scaling vs. Plain GD

Plain GD's first step would be $\Delta \theta^1(x^0) = 0.1(12; 10) = (1.2; 1)$ - a 1.2 : 1 ratio between coordinates, exactly the gradient's own ratio. Adam's actual first step is $(0.1; 0.1)$ - equal in both coordinates, i.e. essentially $2 \cdot \text{sign}(g^0)$ regardless of the 12 vs 10 gradient magnitudes, why? on step 1, $\hat{m}^1 = g^0$ and $\hat{s}^1 = (g^0)^2$ exactly (the bias correction exactly cancels the $(1-\beta)$ factor when $m^0 = s^0 = 0$), so $\text{step}^1 = 2g^0 / \sqrt{(g^0)^2} = 2 \cdot \text{sign}(g^0)$ in every coordinate - Adam's per-coordinate normalization collapses any gradient-magnitude variation on its very first step. By step 2 the two coordinates' moving averages have started to diverge slightly (0.0998 vs. 0.0996), because their gradient histories (not just instantaneous values) differ a little, but the normalization is still visibly pulling the original 12:10 ratio toward 1:1, unlike GD which preserves the ratio exactly at every step.

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